Q.29 What do you understand by phase and phase difference in case of alternating quantities.

Q.30 State and explain Kirchhoff's laws.

Q.31 What is the relation between voltage, current and resistance of an electric circuit.

Q.32 What is meant by self inductance and mutual inductance? Define the units in which each in measured.

Q.33 State any five difference between A.C. and D.C.

Q.34 Define admittance, susceptance and conductance and prove that in a parallel circuit I = VY.

Q.35 Explain two wattmeter method to measure power in a 3-phase unbalanced load.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 State and explain the points to be observed in order to maintain a lead acid cell In good working condition. How would you make sure that the battery has been Completely charged and discharged.

Q.37 A current carrying conductor experiences a force when place in a magnetic field. Name two factors that affect the magnitude of the force.

Q.38 Write short note on any two:

a) 3-phase balanced and unbalanced circuits

b) importance of power factor

c) energy stored in a magnetic field

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Roll No. ....

2nd Sem / Elect, Power Stat. Engg., Elect. & Eltx Engg., Fire Tech & Safety

Subject:- Fundamentals of Electrical Engineering

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following statements is incorrect?

a) Resistance is a passive element

b) inductor is a passive element

c) Current source is passive element

d) voltage source is an active element

Q.2 The specific resistance of a conductor depends upon

a) length of conductor

b) nature of conductor material

c) diameter of conductor

d) all (a), (b) and (c)

Q.3 The specific gravity of acid is checked with the help of

a) hydrometer b) hygrometer

c) lactometer d) cell tester

Q.4 Which of the following is unit of inductance?

a) ohm b) henry

c) ampere turns d) webers/meter

Q.5 The form factor will be

a) 1.0 b) 1.100

c) 1.1 d) 1.187

Q.6 The statement, factor sum of all the e.m.f.'s in any closed circuits is zero is associated with

a) Faraday's law b) ohm's law

c) kirchhoff's law d) coulomb's law

Q.7 In two wattmeter method of measuring power in balance 3-phase circuit, if the readings of two wattmeters are equal but opposite in sign, then the power factor of the load is

a) Unity b) 1/2

c) zero d) 0.5

Q.8 The power factor of series resonant circuit is

a) zero b) unity

c) 0.5 d) always lagging

Q.9 Average emf of nickel & cadmium cell is

a) 1.0 V b) 1.2 V

c) 1.8 V d) 2.0 V

Q.10 Killowatt-hour (kwh) is a unit of

a) current b) power

c) energy d) resistance

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What are primary cell?

Q.12 Draw delta connection.

Q.13 What is EMF?

Q.14 Frequency of DC IS_________.

Q.15 The resistance of a magnetic circuit is similar to_______ of an electric circuit.

Q.16 State Lenz's law.

Q.17 Power factor = active power/________?

Q.18 A wattmeter is used to measured_________

Q.19 Define magnetic flux?

Q.20 The Thevenin voltage is the_________

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 How do you defined the resistance of a conductor? What are the factors upon which it depends?

Q.22 State and explain Norton's theorem.

Q.23 What precautions would you observe in maintenance of a battery?

Q.24 What is the similarity between an electric circuit and a magnetic circuit.

Q.25 State Faraday's laws of electromagnetic induction. Mention two practical applications of this law.

Q.26 In an a.c system, define the following:

a) amplitude b) cycle

c) time period d) frequency

e) phase difference

Q.27 Prove that in a pure resistive circuit, the power consumed by it is $E_{r.m.s} * I_{r.m.s}$

Q.28 Compare star and delta systems of three phase connections.